



UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO

Facultad de Ingeniería

Procesamiento Digital de Señales

Tarea 1

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Tarea 1

DSP Exercise Set: Signals & time-Domain Analysis.

Integrantes:

- ▷ Jaramillo Rodriguez Leslie Citlalli
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Theory Questions

① Define the difference between energy signals and power signals. Give one real-world example of each.

La señal de energía posee energía total finita ($0 < E < \infty$) y potencia promedio igual a 0 ($P=0$). Suele ser transitoria o decaer con el tiempo, un ejemplo es un rayo.

Por otro lado, la señal de potencia posee energía total infinita ($E \rightarrow \infty$) y potencia promedio finita y no nula ($0 < P < \infty$), además suele ser continua o periódica, un ejemplo es la corriente alterna de la red eléctrica.

② State the Nyquist sampling criterion and explain what happens if it is not satisfied

Criterio de Nyquist: La frecuencia de muestreo f_s debe ser al menos el doble de la frecuencia máxima de la señal continua ($f_s \geq 2f_{\max}$).

Si no se satisface se produce 'aliasing', es decir, solapamiento espectral y las altas frecuencias invaden las bajas, distorsionando la señal y haciendo imposible su reconstrucción fiel.

③ Explain in your own words how time reversal and time shifting affect a signal's graph. Why are these operations important in signal analysis?

La inversión temporal es la que refleja la gráfica horizontalmente respecto al eje vertical, mientras que el desplazamiento temporal mueve la gráfica completa a la izquierda (adelanto) o a la derecha (retraso).

Su importancia está en que son cruciales para la alineación de señales, análisis de simetría y operaciones base como la convolución, correlación y filtros adaptados.

Tarea 1

DSP Exercise Set: Signals & Time-Domain Analysis

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1: For each of the following signals, determine whether it is continuous/discrete, periodic/apperiodic, and causal/non-causal:

a) $x(t) = \sin(2\pi 10t)$

- Continua
- $T_0 = \frac{1}{10} = 0.1 \text{ s} \Rightarrow \therefore$ es periódica
- No causal ($t < 0$)

b) $x[n] = u[n - 3]$

- Discreta
- Aperiódica
- Causal

c) $x(t) = e^{-t} u(t)$

- Continua
- Aperiódica
- Causal

2: Is the signal $x[n] = \cos\left(\left(\frac{5\pi}{6}\right)n\right)$ periodic? If yes, find its fundamental period.

$$\omega_0 = \frac{5\pi}{6} \quad \frac{\omega_0}{2\pi} = \frac{K}{N} \Rightarrow \frac{\frac{5\pi}{6}}{2\pi} = \frac{5\pi}{12\pi} = \frac{5}{12}$$

$\therefore N = 12$ y es periódica.

3: A continuous-time signal $x(t) = \cos(200\pi t)$ is sampled at $f_s = 150 \text{ Hz}$

- What is the Nyquist rate?

$$\omega = 200\pi$$

$$f = \frac{200\pi}{2\pi} = 100 \text{ Hz}$$

$$f_{\text{Nyquist}} = 2f = 2(100 \text{ Hz}) = 200 \text{ Hz}$$

- Will aliasing occur? If so, what is the apparent frequency after sampling?

$$\text{SI, porque } f_s = 150 \text{ Hz} < 200 \text{ Hz}$$

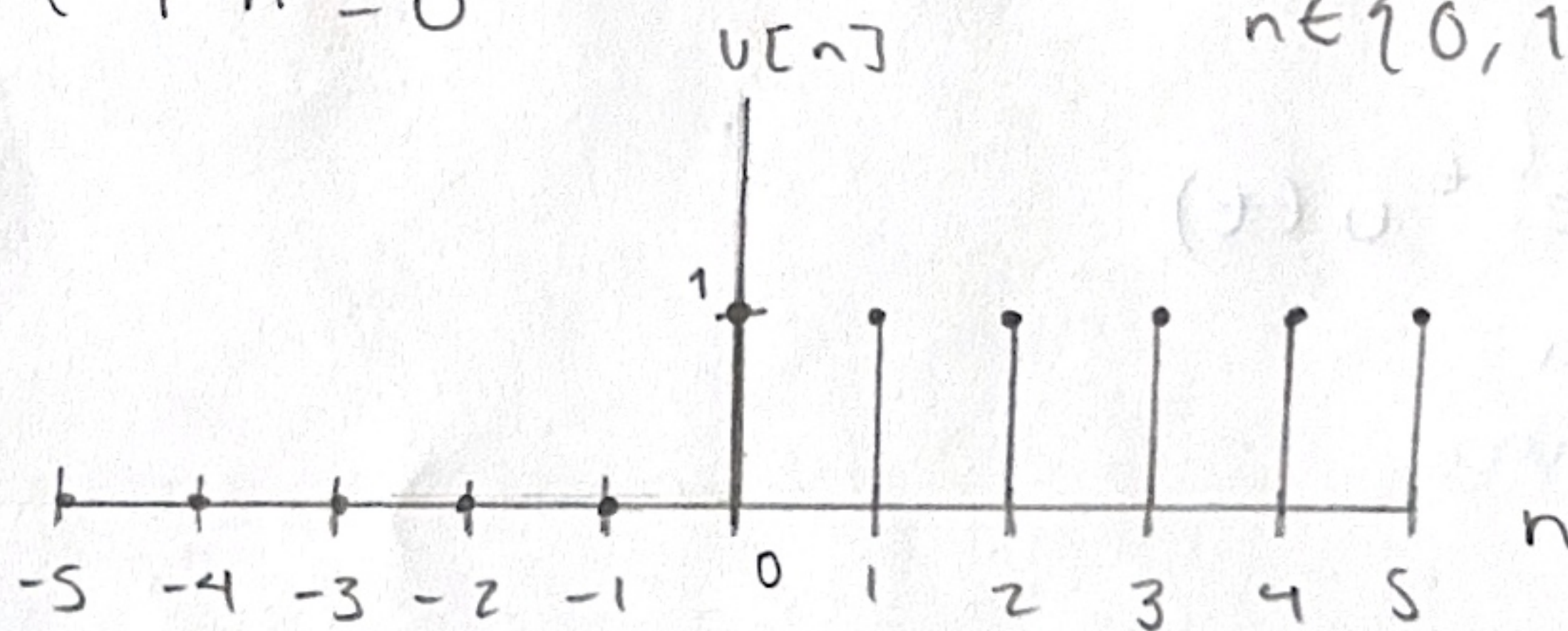
$$f = |f - f_s| = |100 - 150| = 50 \text{ Hz}$$

4: Sketch the following signals for $-5 \leq n \leq 5$:

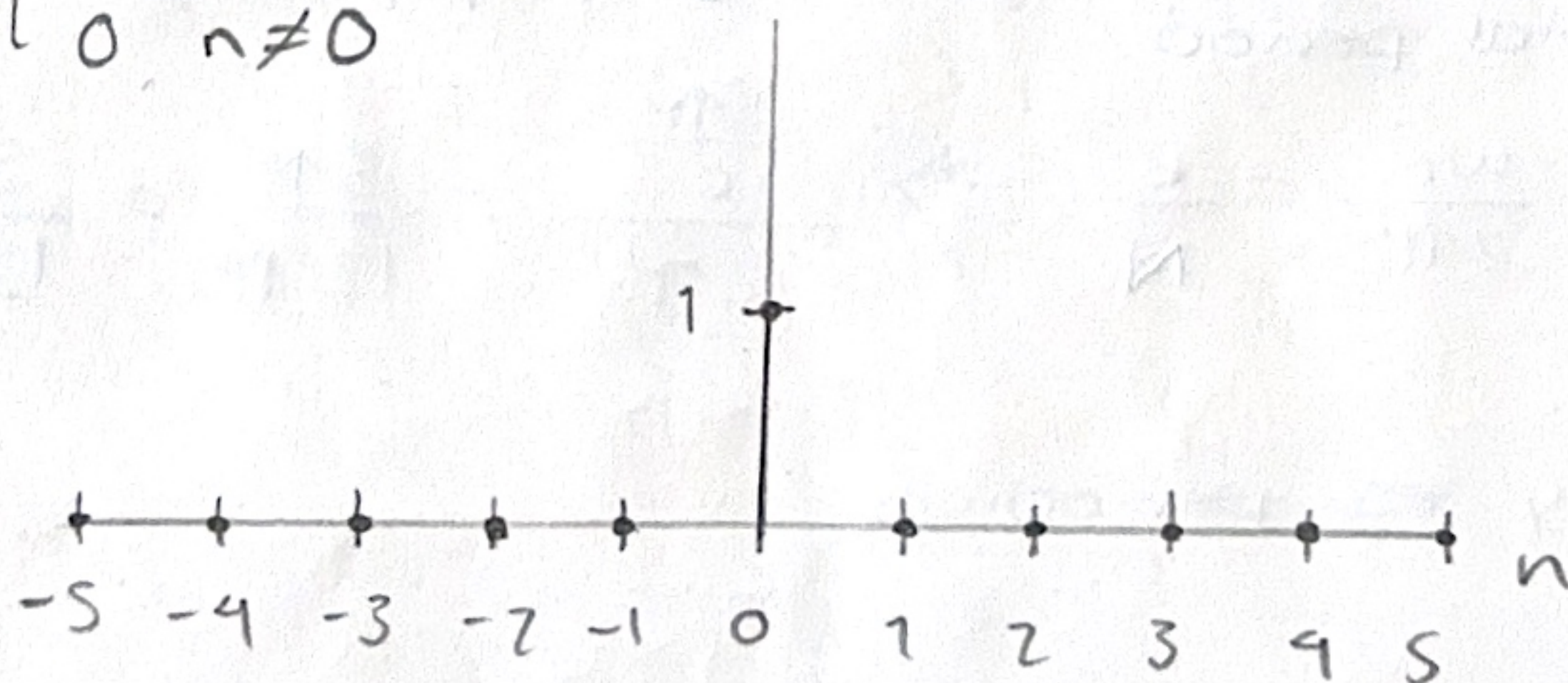
$$a) u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

$$n \in \{-5, -4, -3, -2, -1\} = 0$$

$$n \in \{0, 1, 2, 3, 4, 5\} = 1$$



$$b) \delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



$$c) r[n] = n \cdot u[n] \quad u[n] = \begin{cases} 0 & \text{para } n < 0 \\ 1 & \text{para } n \geq 0 \end{cases}$$

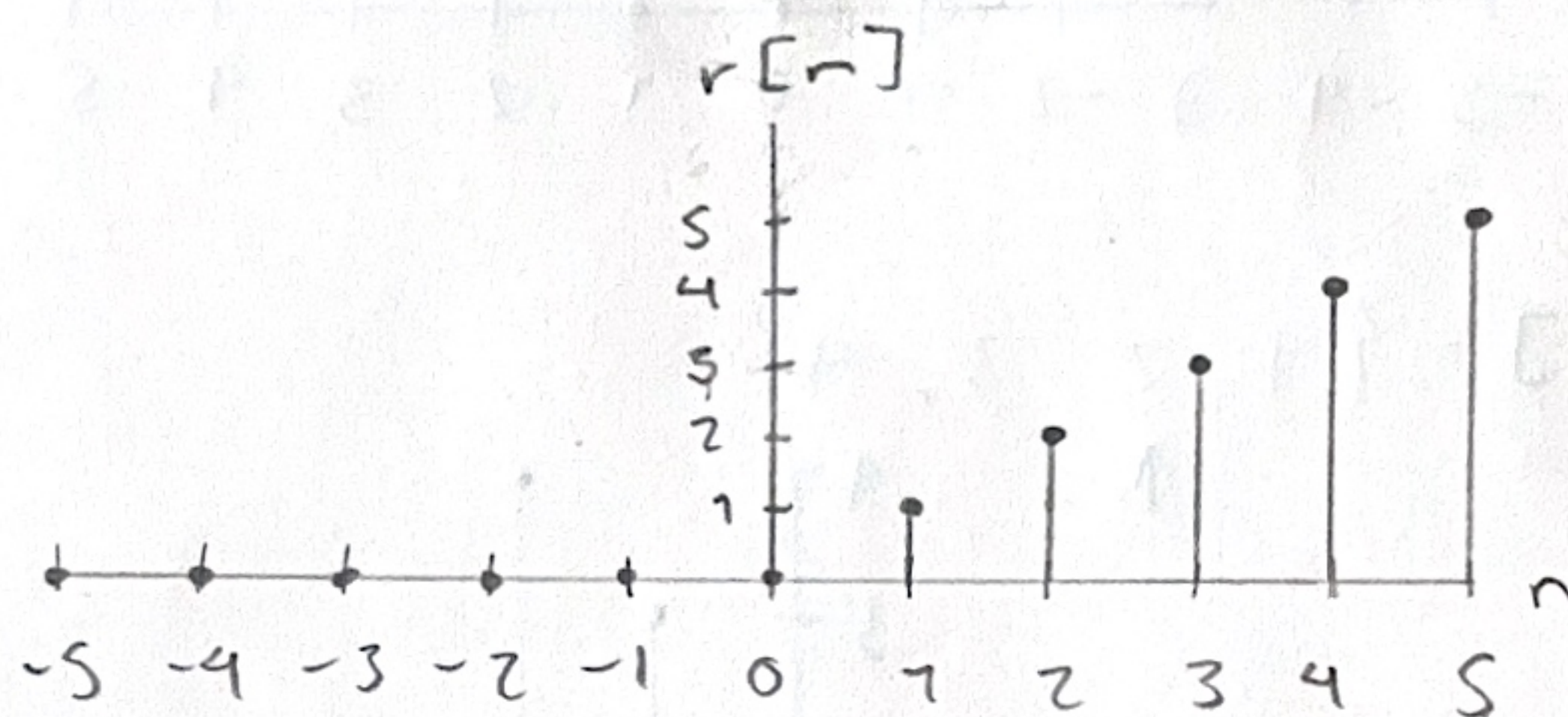
$$r[n] = 0 \quad n < 0$$

$$r[n] = n \quad n \geq 0$$

$$\therefore n \in \{-5, -4, -3, -2, -1\} \Rightarrow 0$$

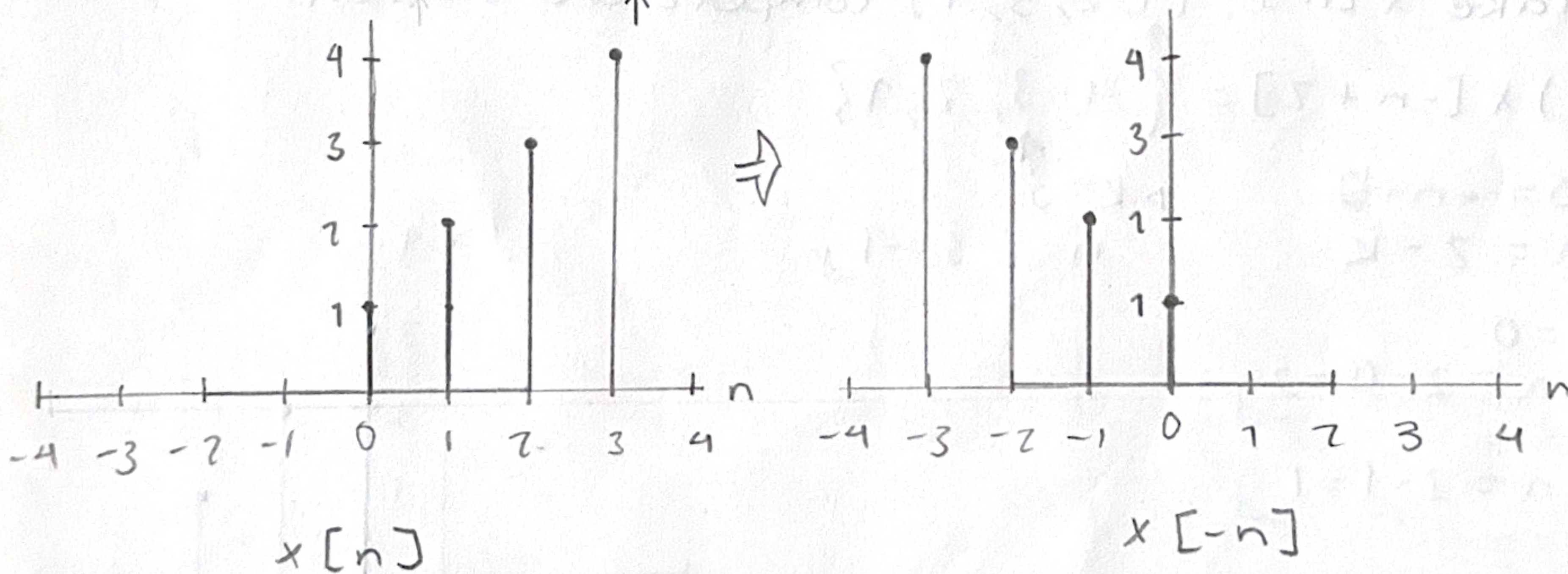
$$n = 0 \Rightarrow 0$$

$$n = 1 \Rightarrow 1, n = 2 \Rightarrow 2, \dots, n = 5 \Rightarrow 5.$$



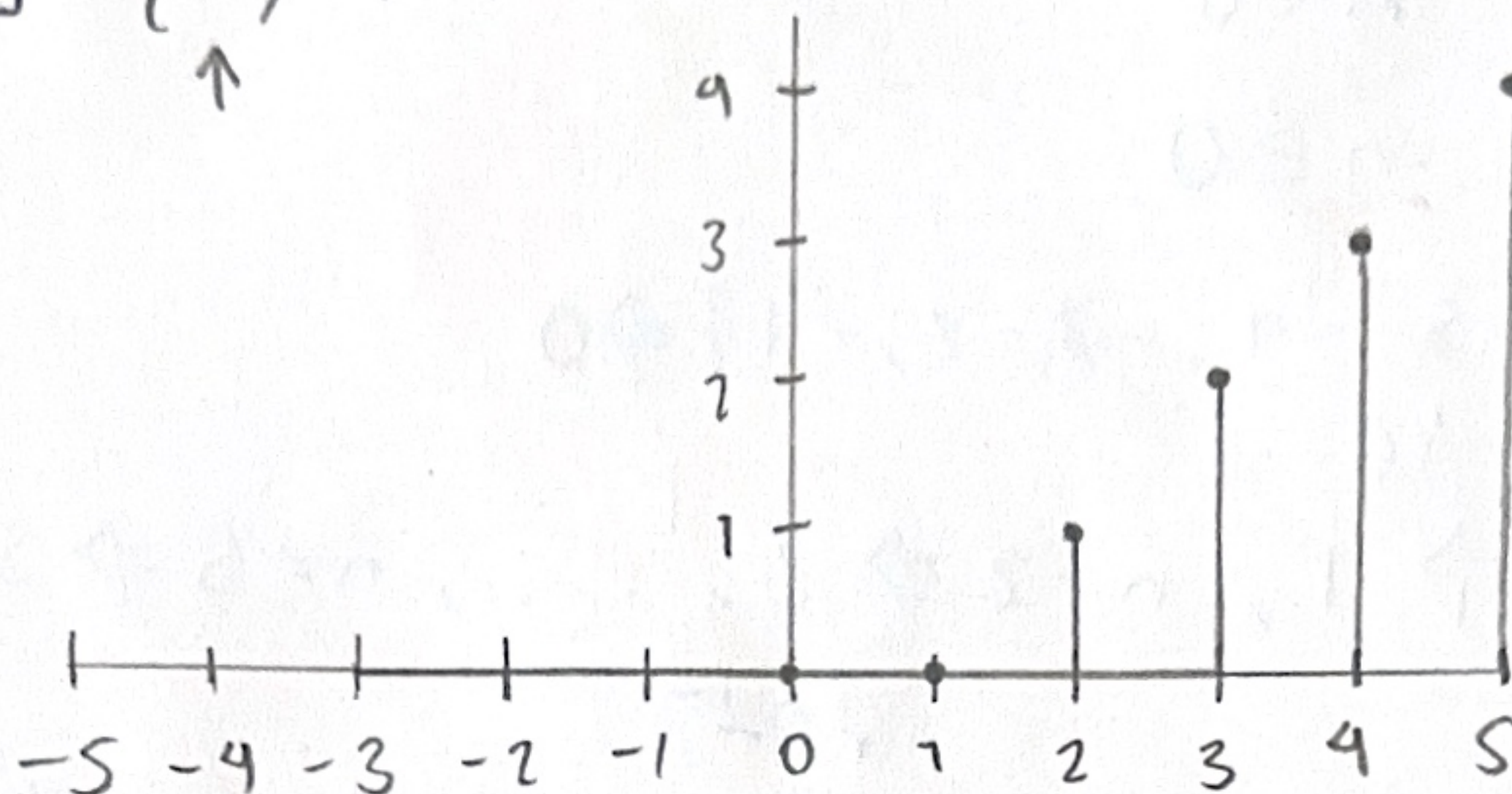
S: Given $x[n] = \{1, 2, 3, 4\}$ for $n = 0, 1, 2, 3$, compute and sketch $x[-n]$.

$$x[-n] = \{4, 3, 2, 1\}$$

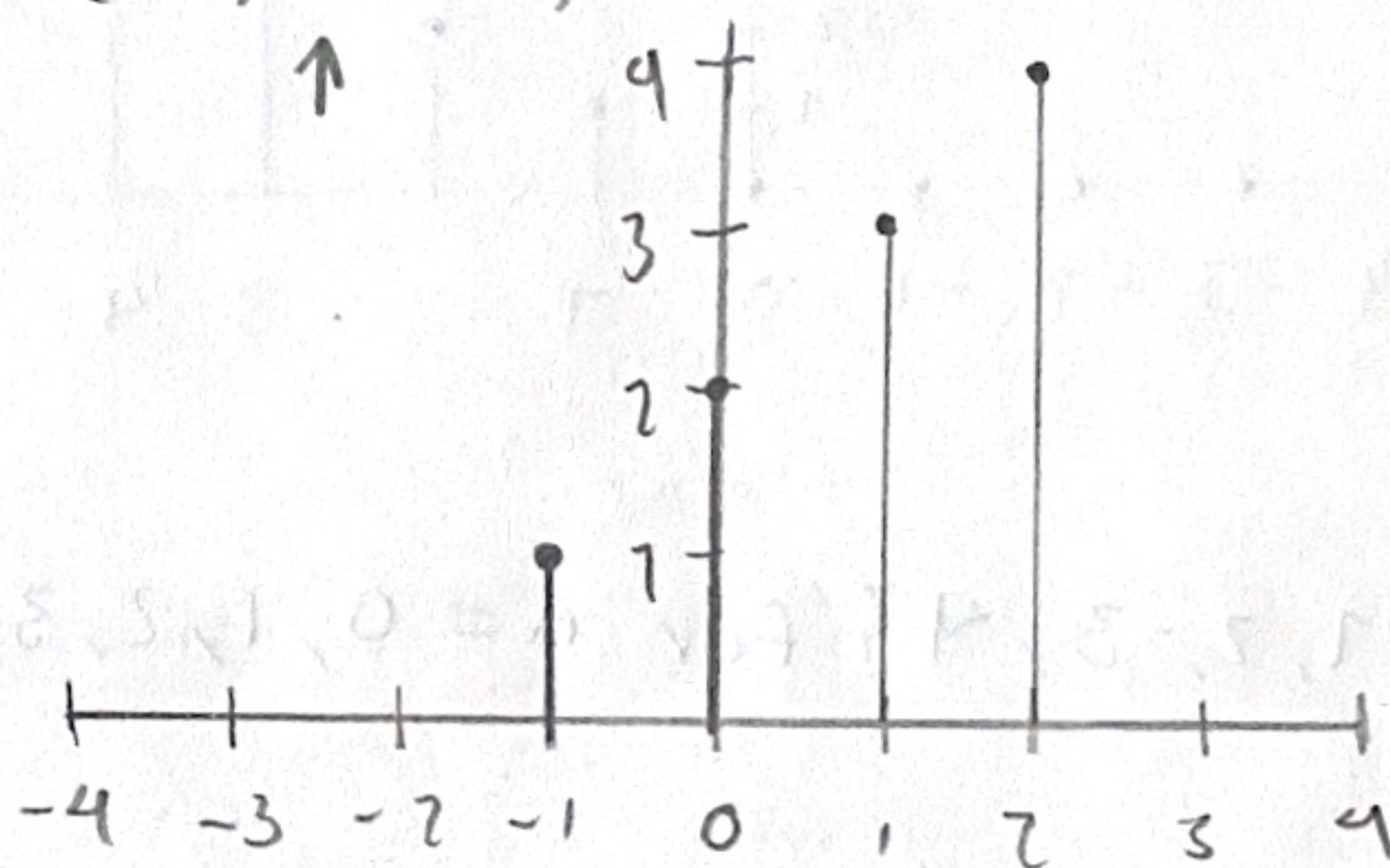


6- For the same signal $x[n] = \{1, 2, 3, 4\}$, compute and sketch:

a) $x[n-2] = \{0, 0, 1, 2, 3, 4\}$



b) $x[n+1] = \{1, 2, 3, 4\}$



7- Take $x[n] = \{1, 2, 3, 4\}$ compute and sketch.

a) $x[-n+2] = \{4, 3, 2, 1\}$

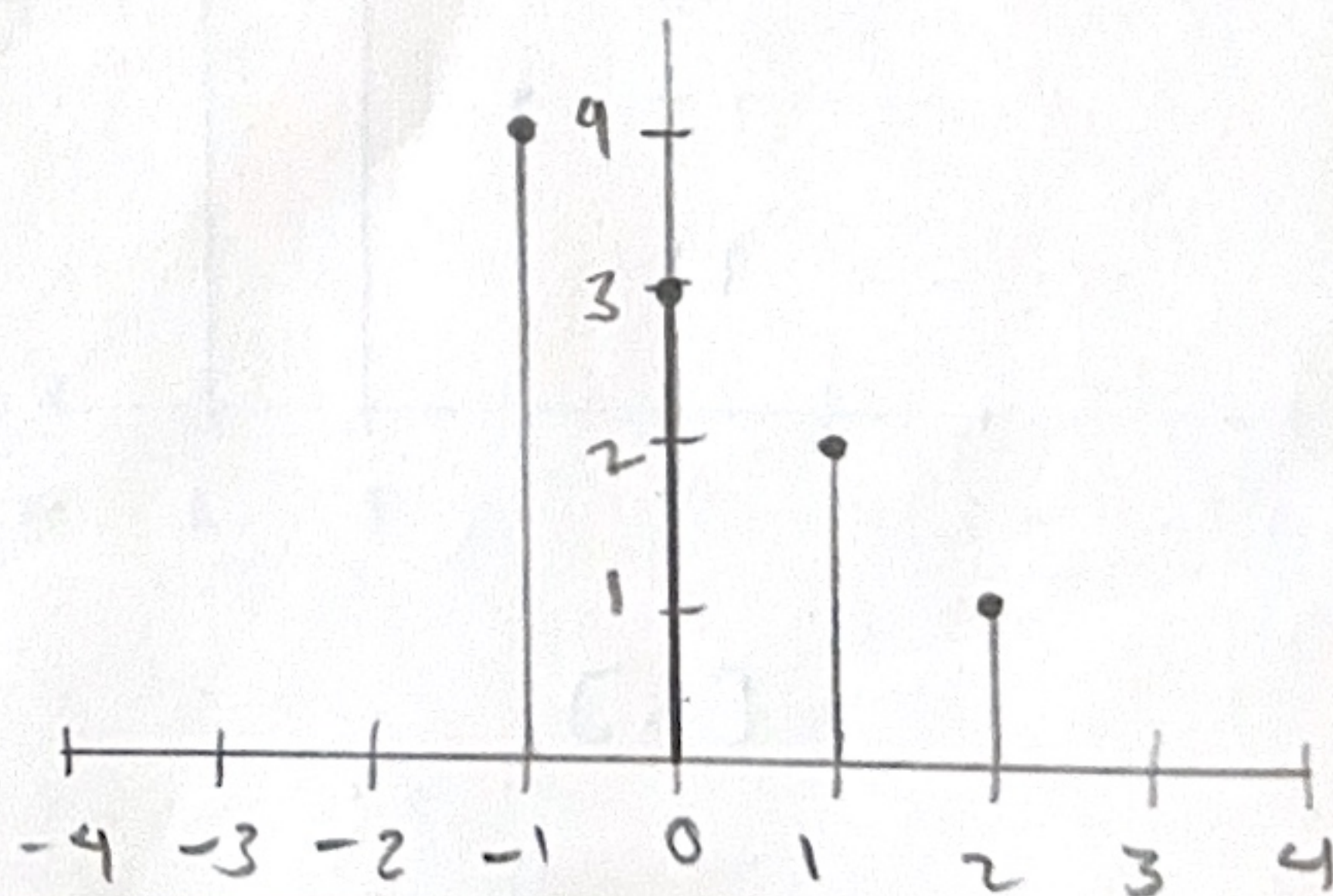
$k = -n + 2$
 $n = 2 - k$

$\triangleright k = 3$
 $n = 2 - 3 = -1$

$\triangleright k = 0$
 $n = 2 - 0 = 2$

$\triangleright k = 1$
 $n = 2 - 1 = 1$

$\triangleright k = 2$
 $n = 2 - 2 = 0$



$$b) x[3-n] = \{ \overset{\downarrow}{4}, 3, 2, 1 \}$$

$$x[n] = \{ \overset{0}{1}, \overset{1}{2}, \overset{2}{3}, \overset{3}{4} \}$$

$$k = 3 - n$$

$$n = 3 - k$$

$$\triangleright k = 0$$

$$n = 3 - 0 = 3$$

$$\triangleright k = 1$$

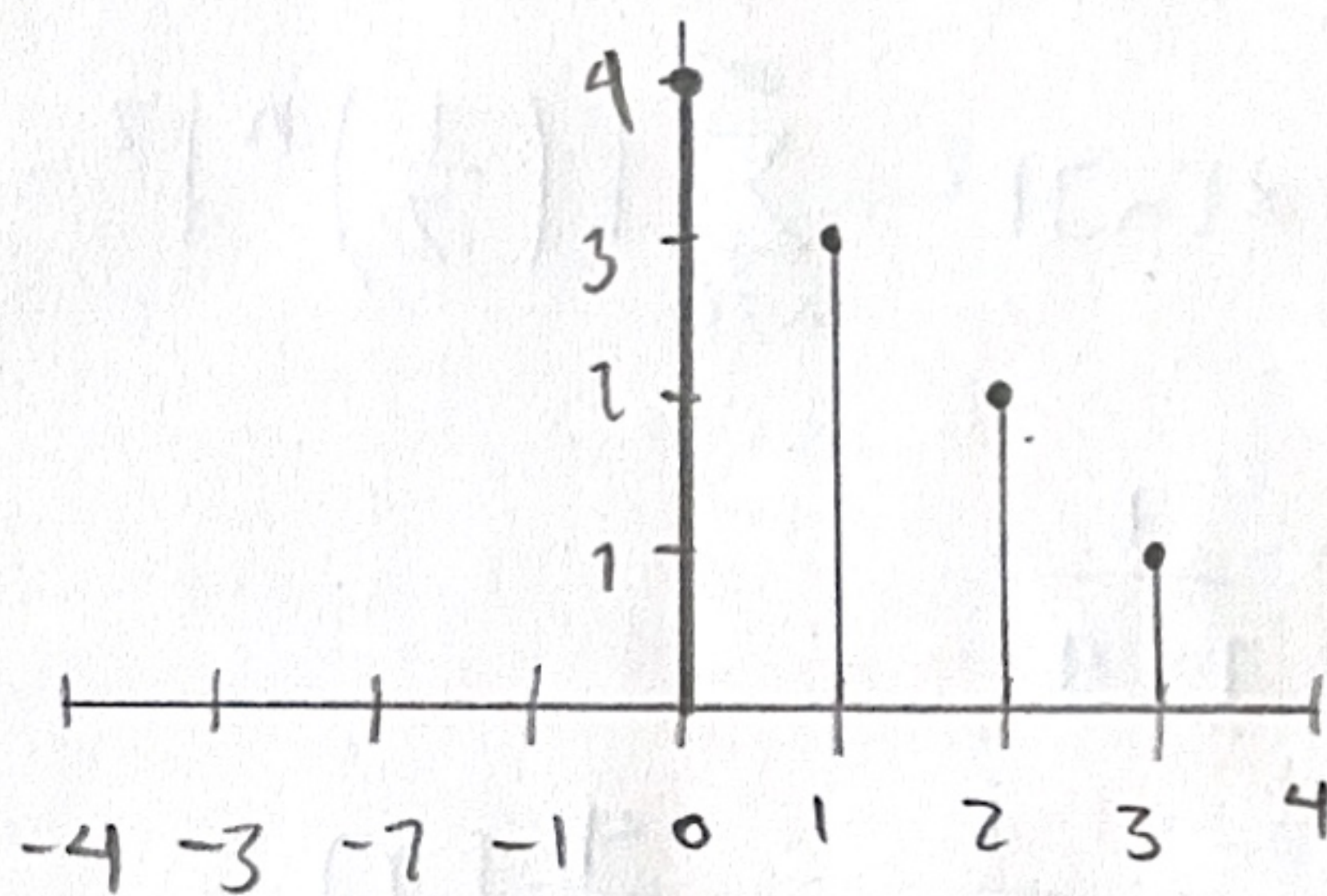
$$n = 3 - 1 = 2$$

$$\triangleright k = 2$$

$$n = 3 - 2 = 1$$

$$\triangleright k = 3$$

$$n = 3 - 3 = 0$$



8- Determine whether the following signals are energy signals or power signals:

$$a) x[n] = \cos\left(\left(\frac{\pi}{4}\right)n\right)$$

$$\omega_0 = \frac{\pi}{4}$$

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \Rightarrow \frac{\frac{\pi}{4}}{2\pi} = \frac{\pi}{8\pi} = \frac{1}{8} \therefore N = 8$$

Es una señal periódica no nula $(-\infty \text{ a } \infty) \therefore E = \infty$
Es una señal de potencia

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{8} \sum_{n=0}^7 \cos^2\left(\frac{\pi}{4}n\right)$$

$$P = \frac{1}{8} \sum_{n=0}^7 \left(\frac{1}{2} + \frac{1}{2} \cos\left(\frac{\pi}{2}n\right) \right) = \frac{1}{8} \left(\sum_{n=0}^7 \frac{1}{2} + \frac{1}{2} \sum_{n=0}^7 \cos\left(\frac{\pi}{2}n\right) \right)$$

$$P = \frac{1}{8} \left(8 \left(\frac{1}{2} \right) + 0 \right) = \frac{1}{2} = .5 \text{ [W]} \quad \checkmark$$

$$b) x[n] = \left(\frac{1}{2} \right)^n u[n]$$

Al decaer hacia 0 conforme $n \rightarrow \infty$, la energía es finita, entonces es una señal de energía.

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=0}^{\infty} \left| \left(\frac{1}{2} \right)^n \right|^2 = \sum_{n=0}^{\infty} \left(\frac{1}{4} \right)^n$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

$$E = \frac{1}{1 - \frac{1}{4}} = \frac{1}{\frac{3}{4}} = \frac{4}{3} \text{ [J]} \quad \checkmark$$

9: Given $x[n] = (.9)^n u[n]$:

- Is this an energy or power signal?
- Compute its total energy.

$|.9| < 1 \therefore$ es una señal aperiódica, que decae exponencialmente, por lo que es una señal de energía.

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \sum_{n=0}^{\infty} (.9^n)^2 = \sum_{n=0}^{\infty} (.9^2)^n = \sum_{n=0}^{\infty} (.81)^n$$

$$E = \frac{1}{1 - .81} = \frac{1}{.19} = 5.263 \text{ [J]} \quad \checkmark$$



10. A signal $x[n] = \{1, -1, 2, -2\}$ for $n = 0, 1, 2, 3$.

- Plot the original signal.
- Apply time reversal $\rightarrow$ shift right by 2 samples $\rightarrow$ scale by factor 3.
- Write the resulting signal explicitly and sketch it.

1- $x_1[n] = x[-n]$

$$x[0] = x[0] = 1$$

$$x[-1] = x[1] = -1$$

$$x[-2] = x[2] = 2$$

$$x[-3] = x[3] = -2$$

$$\left. \begin{array}{l} x[0] = 1 \\ x[-1] = -1 \\ x[-2] = 2 \\ x[-3] = -2 \end{array} \right\} \{-2, 2, -1, 1\}$$

↑

2- $x_2[n] = x_1[n-2] = x[-(n-2)] = x[-n+2]$

$$n = -3 + 2 = -1 \Rightarrow x_2[-1] = -2$$

$$n = -2 + 2 = 0 \Rightarrow x_2[0] = 2$$

$$n = -1 + 2 = 1 \Rightarrow x_2[1] = -1$$

$$n = 0 + 2 = 2 \Rightarrow x_2[2] = 1$$

$$\left. \begin{array}{l} x_2[-1] = -2 \\ x_2[0] = 2 \\ x_2[1] = -1 \\ x_2[2] = 1 \end{array} \right\} \{-2, 2, -1, 1\}$$

↑

3- $x_3[n] = 3(x_2[n])$

$$x_3[-1] = 3(-2) = -6$$

$$x_3[0] = 3(2) = 6$$

$$x_3[1] = 3(-1) = -3$$

$$x_3[2] = 3(1) = 3$$

$$\left. \begin{array}{l} x_3[-1] = -6 \\ x_3[0] = 6 \\ x_3[1] = -3 \\ x_3[2] = 3 \end{array} \right\} \{-6, 6, -3, 3\} = x_3[n]$$

↑

Entonces : $x_3[n] = 3x[-n+2] = \begin{cases} -6, & n = -1 \\ 6, & n = 0 \\ -3, & n = 1 \\ 3, & n = 2 \\ 0, & \text{otro caso} \end{cases}$ ✓

